Supplementary figures for Chapter 5

Large community simulation of the following model:

$$\dot{n}_i = r_i n_i \left(1 - \gamma_i a_i - (1-\gamma_i) n_i - \sum_{j(\neq i)} \kappa_{ij} n_j \right) + r_i \lambda_i, \quad i = 1, \ldots, S, \tag{1}$$

$$\dot{a}_i = n_i - \delta_i a_i, \quad i = 1, \ldots, S. \tag{2}$$

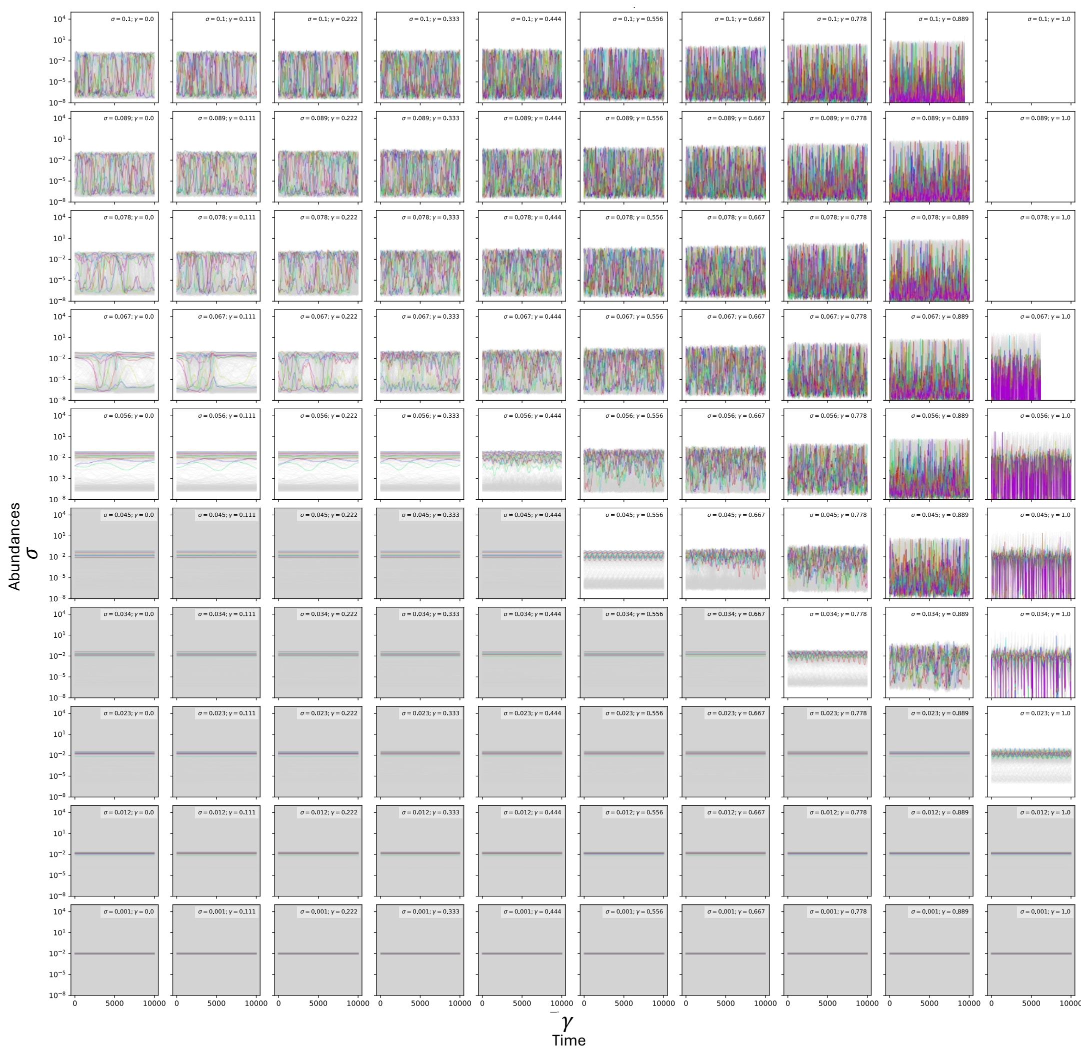

Figure 1: Simulations of the abundances changing σ and γ. The other parameters are fixed, with $\delta = 0.01$, $\mu = 0.2$ and $\lambda = 1e - 8$.

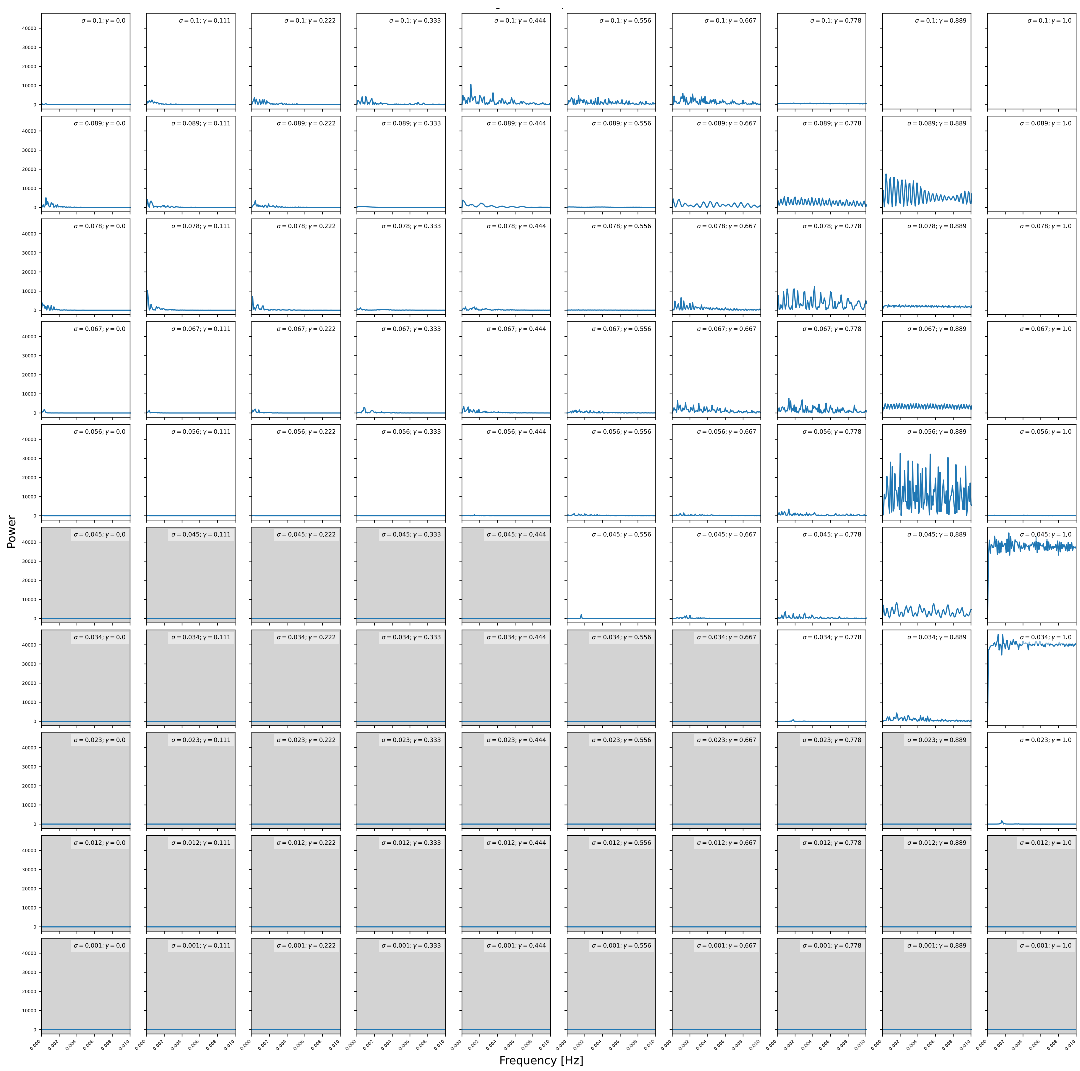

Figure 2: Power spectrum of one random species changing σ and γ. The other parameters are fixed, with $\delta = 0.01$, $\mu = 0.2$ and $\lambda = 1e - 8$

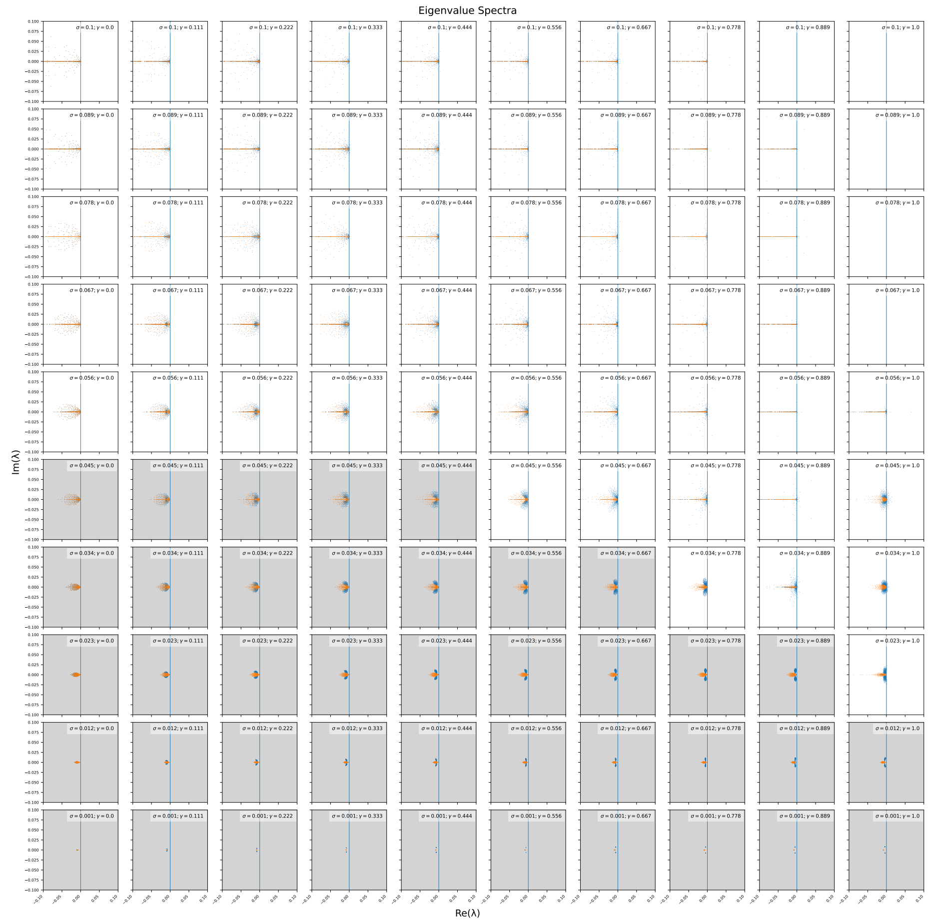

Figure 3: The eigenvalue spectrum of the Jacobian of (1) evaluated at time T, for varying σ and γ. The other parameters are fixed, with $\delta = 0.01$ and $\mu = 0.2$. The orange dots represents the gLV spectrum while the blue dots represents the spectrum of the model with autotoxins. In the first row, when $\gamma = 0$ they are the same.